

1. For the case of quantum harmonic oscillator the potential is given by $V(x) = \frac{1}{2}m\omega^2 x^2$, show that the time-independent Schrodinger equation for the harmonic oscillator can be written as $\frac{1}{2m}[p^2 + (m\omega x)^2]\psi = E\psi$
2. Define the ladder operator. Show that the Hamiltonian of quantum harmonic oscillator can be written as $H = \hbar\omega(a_- a_+ - \frac{1}{2})$
3. Prove that if ψ satisfies the Schrodinger equation with energy E , then $a_+ \psi$ satisfies the Schrodinger equation with energy $(E + \hbar\omega)$.
4. Prove that if ψ satisfies the Schrodinger equation with energy E , then $a_- \psi$ satisfies the Schrodinger equation with energy $(E - \hbar\omega)$.
5. What is the commutation relation? Show that $[x, p] = i\hbar$
6. What is the commutation relation? Show that $[a_-, a_+] = 1$
7. Define bound state and scattering state for the delta function potential.
8. Show that for delta function potential the scattering state solution of the time independent schrodinger equation can give the reflection and transmission coefficient as

$$R = \frac{1}{1+(2\hbar^2 \dots\dots\dots)} \text{ and } T = \frac{1}{1+(m\alpha^2 \dots\dots\dots)}$$
9. Show that for delta function potential, the bound state solution of the time-independent equation can be written as $\psi = \frac{\sqrt{m\alpha}}{\hbar} e^{-m \dots\dots\dots}$
10. Write down the form of Laplacian in cartesian coordinates and spherical polar coordinates system.
11. Write down the Schrodinger equation in the polar coordinate system. Derive the angular and radial equation by using separation of variable technique from the Schrodinger equation.
12. Solve the angular equation to find the spherical harmonics.
13. Show that the effective potential of the radial equation can be written as

$$V_{eff} = V + \hbar^2/2m (l(l + 1))/r^2$$
14. Solve the radial equation to find the energy and wave function for infinite 3D square well potential.